Parametric and Semiparametric Copula Model Selection

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Sklar's Theorem

- A function $C : [0,1]^d \to [0,1]$, is called a **copula** if it is a cumulative distribution function (cdf) with uniform marginals on $[0,1]$.
- **Sklar's Theorem**: Assume that F is the cdf of an **absolutely continuous distribution**. Then there exists a **unique** copula C such that
$$F(x_1, \ldots, x_d) = C(F_1(x_1), \ldots, F_d(x_d)),$$
where $F_1, \ldots, F_d$ are the marginal cdfs of F.
- Sklar's theorem allows us to separate **the dependence structure** from **the structure of the marginal distributions**.
- Since we consider only absolutely continuous distributions, densities can be derived from Sklar's theorem as
$$f(x_1, \ldots, x_d) = c(F_1(x_1), \ldots, F_d(x_d)) \prod_{l=1}^{d} f_l(x_l).$$
- We will be interested in selecting the most appropriate copula c.

Fully Parametric Approach

- Assume that the copula C can be parametrized by a real vector $\boldsymbol{\theta}$, and each marginal cdf F_l can be parametrized by a real vector $\boldsymbol{\gamma}_l$ for $l = 1, \ldots, d$. Then the joint density can be written as

$$f(x_1, \ldots, x_d; \boldsymbol{\theta}, \boldsymbol{\gamma}) = c(F_1(x_1; \boldsymbol{\gamma}_1), \ldots, F_d(x_d; \boldsymbol{\gamma}_d); \boldsymbol{\theta}) \prod_{l=1}^{d} f_l(x_l; \boldsymbol{\gamma}_l),$$

  where $\boldsymbol{\gamma} = (\boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_d)$.

- In statistics, we observe $\mathcal{X}_n = \{\boldsymbol{X}_i\}_{i=1}^{n} \overset{\text{i.i.d}}{\sim} f^\circ$, where f° denotes the unknown data-generating density.
- Let $B \in \mathbb{N}$ denote the number of parametric families under consideration. For each $b = 1, \ldots, B$, define

$$\mathcal{F}_b = \left\{ f(\cdot; \boldsymbol{\theta}_b, \boldsymbol{\gamma}_{b,1}, \ldots, \boldsymbol{\gamma}_{b,d}) : (\boldsymbol{\theta}_b, \boldsymbol{\gamma}_{b,1}, \ldots, \boldsymbol{\gamma}_{b,d}) \in \Theta_b \times \Gamma_{b,1} \ldots \times \Gamma_{b,d} \right\}.$$

- Our goal is to **rank the fitted models** to identify the most suitable one.

MLE-Based Information Criteria

- By fitting each family $\mathcal{F}_b$ to the data $\mathcal{X}_n$, one can obtain the MLEs
$$\left(\widehat{\boldsymbol{\theta}}_b, \widehat{\gamma}_{b,1}, \ldots, \widehat{\gamma}_{b,d}\right).$$
- If $f^\circ \in \mathcal{F}_b$, we compute the Akaike Information Criterion:
$$\mathrm{AIC}_{\mathrm{MLE}}(b) = 2 \cdot \left[\ell\left(\widehat{\boldsymbol{\theta}}_b, \widehat{\gamma}_{b,1}, \ldots, \widehat{\gamma}_{b,d}\right) - \dim(\boldsymbol{\theta}_b, \gamma_{b,1}, \ldots, \gamma_{b,d})\right].$$
- If $f^\circ \notin \mathcal{F}_b$, we use the Takeuchi Information Criterion:
$$\mathrm{TIC}_{\mathrm{MLE}}(b) = 2 \cdot \left[\ell\left(\widehat{\boldsymbol{\theta}}_b, \widehat{\gamma}_{b,1}, \ldots, \widehat{\gamma}_{b,d}\right) - \widehat{\mathrm{bias}}_{\mathrm{MLE}}(b)\right].$$
- **Problem**: The joint estimation of $(\boldsymbol{\theta}, \gamma_1, \ldots, \gamma_d)$ is often **computationally expensive** fot high-dimensional data.

Two-Stage MLE

The **joint log-likelihood function** for the sample is given by:

$$\ell(\boldsymbol{\eta}) = \sum_{i=1}^{n} \log c\left(F_1(X_{i,1};\gamma_1),\dots,F_d(X_{i,d};\gamma_d);\boldsymbol{\theta}\right) + \sum_{i=1}^{n}\sum_{l=1}^{d} \log f_l(X_{i,l};\gamma_l),$$

where $\boldsymbol{\eta} = (\boldsymbol{\theta},\gamma_1,\dots,\gamma_d)$.

Stage 1 (Marginal Estimation): For each $l = 1,\dots,d$, obtain the marginal maximum likelihood estimate $\widetilde{\gamma}_l$ by maximizing the marginal log-likelihood with respect to γ_l:

$$\widetilde{\gamma}_l = \arg\max_{\gamma_l} \sum_{i=1}^{n} \log f_l(X_{i,l};\gamma_l).$$

Two-Stage MLE

Stage 2 (Copula Estimation): Plug in $\widetilde{\gamma} = (\widetilde{\gamma}_1, \ldots, \widetilde{\gamma}_d)$ from Stage 1, to get

$$\ell(\boldsymbol{\theta}, \widetilde{\gamma}) = \sum_{i=1}^{n} \log c\left(F_1\left(X_{i,1}; \widetilde{\gamma}_1\right), \ldots, F_d\left(X_{i,d}; \widetilde{\gamma}_d\right); \boldsymbol{\theta}\right) + \sum_{i=1}^{n} \sum_{l=1}^{d} \log f_l(X_{i,l}; \widetilde{\gamma}_l).$$

Then, $\boldsymbol{\theta}$ is estimated by maximizing $\ell(\boldsymbol{\theta}, \widetilde{\gamma})$ with respect to $\boldsymbol{\theta}$, yielding

$$\widetilde{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}, \widetilde{\gamma}).$$

The final two-stage maximum likelihood estimate for the entire model is

$$\widetilde{\boldsymbol{\eta}} = (\widetilde{\boldsymbol{\theta}}, \widetilde{\gamma}).$$

Remark

If one wishes to rank the fitted models using these two-stage estimates $\widetilde{\boldsymbol{\eta}}$, the standard $\mathrm{TIC}_{\mathrm{MLE}}$ formula cannot be applied.

Two-Stage MLE based Information Criteria

- In Ko and Hjort (2019), the copula information criterion $\mathrm{CIC}_{\mathrm{2MLE}}$ was derived for two-stage MLE.
- This criterion is an analogue of the $\mathrm{TIC}_{\mathrm{MLE}}$ in the sense that it **does not assume** that $\mathcal{F}_b$ includes f°:

$$\mathrm{CIC}_{\mathrm{2MLE}}(b) = 2\left[\ell(\widetilde{\boldsymbol{\eta}}_b) - \widetilde{\mathrm{bias}}_{\mathrm{2MLE}}(b)\right].$$

- The authors also showed that if $\mathcal{F}_b$ includes f°, the $\mathrm{CIC}_{\mathrm{2MLE}}$ simply reduces to:

$$\mathrm{AIC}_{\mathrm{2MLE}}(b) = 2\left[\ell(\widetilde{\boldsymbol{\eta}}_b) - \dim(\boldsymbol{\eta}_b)\right].$$

- This means that the naive adaptation of $\mathrm{AIC}_{\mathrm{MLE}}$ to the two-stage MLE is theoretically justified, assuming $f^\circ \in \mathcal{F}_b$.
- To investigate the behavior of the $\mathrm{CIC}_{\mathrm{2MLE}}$ and the $\mathrm{AIC}_{\mathrm{2MLE}}$, we replicate the simulation study presented in Ko and Hjort (2019).

Simulation Objectives

The primary objectives are to:

- Examine the similarity/dissimilarity between $\mathrm{CIC}_{\mathrm{2MLE}}$ and $\mathrm{AIC}_{\mathrm{2MLE}}$;
- Determine if models with higher $\mathrm{CIC}_{\mathrm{2MLE}}$ or $\mathrm{AIC}_{\mathrm{2MLE}}$ values are truly "better" in practice;
- Compare the behavior of the bias-correction term $\widetilde{\mathrm{bias}}_{\mathrm{2MLE}}$ and the naive penalty $\dim(\boldsymbol{\eta})$.

Data-Generating Process

- We assume the true data-generating density f° is a 4-dimensional pdf ($d = 4$).
- **Dependence:** Governed by a Gumbel copula with parameter $\theta^\circ = 3$.
- **Marginal Distributions:**
  - Margin 1: Weibull with $\gamma_1^\circ = 1.5$ (shape) and $\gamma_2^\circ = 4$ (scale).
  - Margin 2: Weibull with $\gamma_3^\circ = 2$ (shape) and $\gamma_4^\circ = 3$ (scale).
  - Margin 3: Gamma with $\gamma_5^\circ = 2$ (shape) and $\gamma_6^\circ = 1$ (rate).
  - Margin 4: Gamma with $\gamma_7^\circ = 3$ (shape) and $\gamma_8^\circ = 1$ (rate).

We generate $n = 1000$ independent observations per simulation run from this true model.

Candidate Copulas and Margins

To evaluate the criteria, we define a set of $B = 324$ fully parametric models based on all possible combinations of the candidates in Table 1.

	Candidates
Copula	Gumbel, Gaussian, Frank, Survival Clayton
Margin 1	Weibull, Gamma, Log-normal
Margin 2	Weibull, Gamma, Normal
Margin 3	Gamma, Weibull, Log-normal
Margin 4	Gamma, Weibull, Log-normal

Table 1: List of the candidate copulas and margins.

Evaluation Metric: Joint Tail Probability

Motivation: Theory vs. Practice

- CIC_{2MLE} minimizes estimated **Kullback-Leibler divergence**.
- While theoretically sounds good, does it really identify models that are useful in practice (e.g., in finance or hydrology)?

Practical Utility Metric: Multivariate Extremes

- We focus on the probability of simultaneous extreme events:

$$\mathrm{TP}^\circ = \mathrm{P}\left(X_1 > q_{0.7}^{(1)}, \ldots, X_4 > q_{0.7}^{(4)}\right) = \mathrm{P}\Big(F_1^\circ(X_1) > 0.7, \ldots, F_4^\circ(X_4) > 0.7\Big), \qquad (1)$$

where $q_{0.7}^{(l)} = (F_l^\circ)^{-1}(0.7)$ are the true 0.7-quantiles.

Simulation Setup

- For each of the **324 candidate models**, we estimate this probability.
- **Performance Measure:** Mean Squared Error (MSE) relative to the true probability (1) across 1000 Monte-Carlo replications.

Estimation of the Joint Tail Probability

- For each of the 324 candidate models, we obtain the two-stage MLEs $\left(\widetilde{\boldsymbol{\theta}}^{(m)}, \widetilde{\gamma}_1^{(m)}, \widetilde{\gamma}_2^{(m)}, \widetilde{\gamma}_3^{(m)}, \widetilde{\gamma}_4^{(m)}\right)$ in each Monte Carlo replication $m = 1, \ldots, M$.
- We map the true thresholds $q_{0.7}^{(l)}$ to the unit interval $[0, 1]$ using the estimated marginal cdfs:

$$\widetilde{u}_l^{(m)} = F_l\left(q_{0.7}^{(l)}; \widetilde{\gamma}_l^{(m)}\right) \quad \text{for } l = 1, \ldots, 4.$$

- The joint tail probability (1) is then estimated as:

$$\widetilde{\mathrm{TP}}_m = \int_{\widetilde{u}_1^{(m)}}^1 \int_{\widetilde{u}_2^{(m)}}^1 \int_{\widetilde{u}_3^{(m)}}^1 \int_{\widetilde{u}_4^{(m)}}^1 c\left(u_1, u_2, u_3, u_4; \widetilde{\boldsymbol{\theta}}^{(m)}\right) du_1\, du_2\, du_3\, du_4.$$

- Finally, we calculate the MSE across $M = 1000$ replications relative to the true joint tail probability TP°:

$$\mathrm{MSE} = \frac{1}{M} \sum_{m=1}^{M} \left(\widetilde{\mathrm{TP}}_m - \mathrm{TP}^\circ\right)^2.$$

CIC_{2MLE} vs AIC_{2MLE}

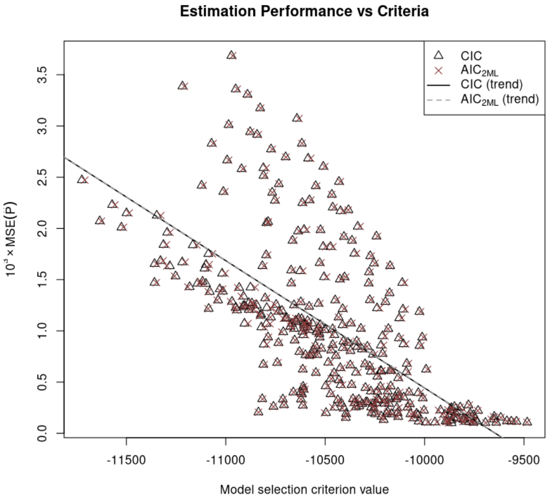

Figure 1: The Mean Squared Error of the estimated joint tail probability (1) versus the average values of the CIC_{2MLE} and the AIC_{2MLE} for all 324 candidate models across 1000 Monte Carlo replications.

Sensitivity of Bias-Correction Terms

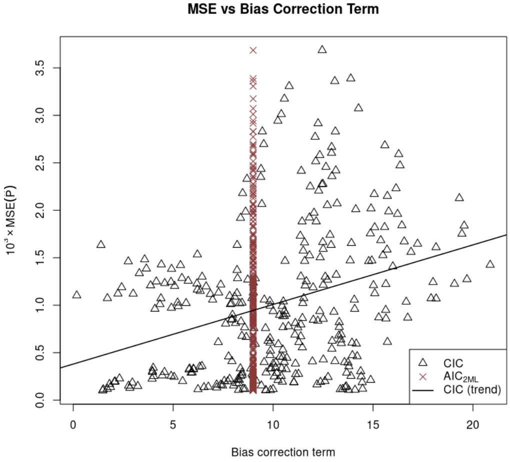

Figure 2: Mean squared error (MSE) of the estimated joint tail probability (1) plotted against the average bias-correction terms: $\widetilde{\text{bias}}_{\text{2MLE}}$ for the CIC_{2MLE} and $\dim(\boldsymbol{\eta})$ for the AIC_{2MLE}. The MSE and average penalties are computed across 1000 Monte Carlo replications.

Summary: Fully Parametric Approach

Key Findings for 2-Stage MLE:

- CIC_{2MLE} penalizes misspecified models more heavily through its complex bias-correction term. However, this conceptual superiority provides little practical advantage for large sample sizes (e.g., $n = 1000$).
- For smaller sample sizes, the empirically estimated $\widetilde{\text{bias}}_{\text{2MLE}}$ could potentially dominate the likelihood term, leading to increased variability in the CIC_{2MLE} performance.

Drawbacks of the fully parametric approach:

- The estimates of the dependence parameters $\boldsymbol{\theta}$ are highly sensitive to the choice of the marginal models.
- Misspecified marginals can severely distort the estimates of the copula parameters $\boldsymbol{\theta}$.

How do we select copula models when the true nature of the marginals is completely unknown?

$\rightarrow$ **Part II: The Semiparametric Approach**

Semiparametric Approach

- Assume that we are only interested in the reliable estimation of the dependence parameters $\boldsymbol{\theta}$, and we don't want to make any parametric assumptions about marginals.
- Instead of fitting models to the **independent observations** $\mathcal{X}_n \subset \mathbb{R}^d$, we fit only copula models to the **dependent pseudo-observations** ${}^p\mathcal{X}_n \subset [0,1]^d$.
- The transformation $\mathcal{X}_n \longmapsto {}^p\mathcal{X}_n$ is defined by the function

$$\boldsymbol{F}_{n,\perp}(x_1,\ldots,x_d) = (F_{n,1}(x_1),\ldots,F_{n,d}(x_d)),$$

  where $F_{n,l}$ is the $\frac{n}{n+1}$-rescaled empirical cdf of the l-th marginal, for $l = 1,\ldots,d$.
- The corresponding pseudo-observations ${}^p\mathcal{X}_n = \{{}^p\boldsymbol{X}_i\}_{i=1}^n$ are then given by ${}^p\boldsymbol{X}_i = \boldsymbol{F}_{n,\perp}(\boldsymbol{X}_i)$, for all $i = 1,\ldots,n$.

Maximum Pseudo-Likelihood Estimator (MPLE)

- The **pseudo-log-likelihood** is then defined as

$$^{p}\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} \log[c(^{p}\boldsymbol{X}_i; \boldsymbol{\theta})].$$

- The **maximum pseudo-likelihood estimator (MPLE)** is given by

$$^{p}\widehat{\boldsymbol{\theta}} = \underset{\boldsymbol{\theta}\in\Theta}{\operatorname{argmax}}\ {}^{p}\ell(\boldsymbol{\theta}).$$

- Genest et al. (1995) showed that this maximizer exhibits the same asymptotic behavior as the classical MLE. Specifically, it is consistent:

$$^{p}\widehat{\boldsymbol{\theta}} \xrightarrow[n\to\infty]{\mathrm{P}} \underset{\boldsymbol{\theta}\in\Theta}{\arg\min} \int_{[0,1]^d} \log\left[\frac{c^{\circ}(\boldsymbol{u})}{c(\boldsymbol{u};\boldsymbol{\theta})}\right] dC^{\circ}(\boldsymbol{u}).$$

Naive Adaptation of AIC to the Semiparametric Case

- Consider $B \in \mathbb{N}$ candidate parametric copula families:

$$\mathcal{C}_b = \{c(\cdot;\boldsymbol{\theta}_b) : \boldsymbol{\theta}_b \in \Theta_b\} \quad \text{for } b = 1, \ldots, B.$$

- Using MPLE, we fit the candidate copula families to our data to identify the most suitable model.
- A **naive adaptation** of AIC to the semiparametric case yields

$$^{p}\mathrm{AIC} = 2 \cdot \left[{}^{p}\ell\left({}^{p}\widehat{\boldsymbol{\theta}}\right) - \dim(\boldsymbol{\theta}) \right].$$

- When using the ^{p}AIC, it was initially hoped that because the MPLE heuristically resembled the MLE, there would be a continuous relationship between the ^{p}AIC and the classical $\mathrm{AIC}_{\mathrm{MLE}}$.

Problems with ^{p}AIC

- ^{p}AIC ignores the noise introduced when transforming data into pseudo-observations.
- Grønneberg and Hjort (2014) adapted classical AIC_{MLE} arguments to the semiparametric setting, deriving an information criterion valid only under **highly restrictive conditions**.
- However, these **conditions are rarely satisfied** by popular copula models.
- Furthermore, they proved a much stronger result: a generally applicable **AIC-like formula does not even exist** for many popular copula models under MPLE.
- **Conclusion:** In the copula context, **when viewing model selection from a loss-function perspective**, standard penalty-based criteria are simply not suitable for pseudo-likelihood estimation.
- Nevertheless, ^{p}AIC remains widely used due to its computational simplicity.

Equivalence Between $\mathrm{TIC}_{\mathrm{MLE}}$ and xv_1

The Classical Parametric Setting (MLE)

- It is well known that $\mathrm{TIC}_{\mathrm{MLE}}$ is asymptotically equivalent to leave-one-out cross-validation xv_1:

$$\mathrm{TIC}_{\mathrm{MLE}} = 2n \cdot \mathrm{xv}_1 + o_P(1), \tag{2}$$

where $\mathrm{xv}_1 = \frac{1}{n}\sum_{i=1}^{n} \log\left[f\left(\boldsymbol{X}_i; \widehat{\boldsymbol{\eta}}_{(-i)}\right)\right]$, and $\widehat{\boldsymbol{\eta}}_{(-i)}$ is the MLE based on the sample $\mathcal{X}_n$ without the i-th observation $\boldsymbol{X}_i$.

The Semiparametric Setting MPLE

- However, in Section 4 of Grønneberg and Hjort (2014), it was shown that in the case of the MPLE, an equivalence analogous to (2) does not hold, which provides a further contrast between MPLE-based and MLE-based estimation.

Leave-One-Out Copula Information Criterion

Motivated by a **prediction perspective**, the leave-one-out cross-validation information criterion is defined as:

$$ {}^{p}\mathrm{xv}_1 = \frac{1}{n} \sum_{i=1}^{n} \log \left[c \left(\boldsymbol{F}_{n,\perp,(-i)}(\boldsymbol{X}_i); {}^{p}\widehat{\boldsymbol{\theta}}_{(-i)} \right) \right], \text{ where} $$

- $\boldsymbol{F}_{n,\perp,(-i)}(x_1, \ldots, x_d) = \left(F_{(-i),1}(x_1), \ldots, F_{(-i),d}(x_d) \right)$, where $F_{(-i),l}$ is the $\frac{n-1}{n}$-rescaled empirical cdf of the l-th marginal, computed from the sample $\mathcal{X}_n$ excluding $\boldsymbol{X}_i$, for $l = 1, \ldots, d$,
- ${}^{p}\widehat{\boldsymbol{\theta}}_{(-i)} = \underset{\boldsymbol{\theta} \in \Theta}{\operatorname{argmax}} \sum_{j \neq i} \log \left[c \left(\boldsymbol{F}_{n,\perp,(-i)}(\boldsymbol{X}_j); \boldsymbol{\theta} \right) \right]$.

Bias-Corrected Approximation of ${}^{p}\mathrm{xv}_1$

However, since computing ${}^{p}\mathrm{xv}_1$ is **computationally expensive**, the authors recommend using its first-order asymptotic approximation, $\mathrm{xv}_{\mathrm{CIC}}$, defined as:

$$\mathrm{xv}_{\mathrm{CIC}} = 2 \cdot \left[{}^{p}\ell \left({}^{p}\widehat{\boldsymbol{\theta}} \right) - \widehat{\mathrm{bias}}_{\mathrm{CIC}} \right],$$

where $\widehat{\mathrm{bias}}_{\mathrm{CIC}}$ is a bias-correction term (see Section 4 of Grønneberg and Hjort (2014) for details).

Paradox: Although ${}^{p}\mathrm{xv}_1$ and $\mathrm{xv}_{\mathrm{CIC}}$ are both **theoretically valid** information criteria, simulation studies (see Jordanger and Tjøstheim (2014)) demonstrate that **neither can outperform** ${}^{p}\mathrm{AIC}$ — a criterion that lacks theoretical justification.

Alternative Cross-Validation Approaches

- In the context of **linear model selection**, Shao (1993) showed that the optimal selection procedure is leave-n_v-out cross-validation, where the **the validation set size** n_v is of the same order as the full sample size n, that is, $n_v/n \to 1$ as $n \to \infty$.
- During the summer of 2025, I attempted to "naively" adapt Shao's approach to the copula setting. The proposed method was still **unable to beat** the well-known ^{p}AIC.
- As an obvious **cross-validation-based** alternative, one can try using K-fold cross-validation with the standard values $K = 5, 10$.
- Are there **other possible choices** for K, and what is the **motivation** behind them?

Choosing K_n: A Data-Driven Approach

Basic properties of leave-one-out cross-validation ${}^p\mathrm{xv}_1$:

1. $n_v = 1$,
2. $\frac{n_v}{n} \underset{n\to\infty}{\longrightarrow} 0$,
3. $K_n = \frac{n}{n_v} = \frac{n}{1} = n$ $\left[{}^p\mathrm{xv}_1 \text{ is a special case of } K_n\text{-fold cross-validation}\right]$.

- Since ${}^p\mathrm{xv}_1$ is a **theoretically justified** information criterion and is a **special case** of ${}^p\mathrm{xv}_{K_n}$, one might wonder how ${}^p\mathrm{xv}_{K_n}$ would perform for values of K_n such that the property (2) still holds.
- In other words, ${}^p\mathrm{xv}_{K_n}$ aims to imitate ${}^p\mathrm{xv}_1$ in the sense that the validation set remains small, while avoiding the overly "conservative" choice $n_v = 1$.
- **The question of interest** is whether ${}^p\mathrm{xv}_{K_n}$ can outperform ${}^p\mathrm{AIC}$.

K_n-Fold Cross-Validation

We randomly partition the training data $\mathcal{X}_n$ into K_n disjoint subsets of equal size $n_v = \frac{n}{K_n}$. Let $\mathcal{I}_k$ denote set of indices belonging to the k-th fold for $k = 1, \ldots, K_n$.

$$^p\mathrm{xv}_{K_n} = \frac{1}{n} \sum_{k=1}^{K_n} \sum_{i \in \mathcal{I}_k} \log \left[c \left(\boldsymbol{F}_{n,\perp,(-\mathcal{I}_k)}(\boldsymbol{X}_i); {}^p\widehat{\boldsymbol{\theta}}_{(-\mathcal{I}_k)} \right) \right], \text{ where}$$

- ${}^p\widehat{\boldsymbol{\theta}}_{(-\mathcal{I}_k)} = \underset{\boldsymbol{\theta} \in \Theta}{\operatorname{argmax}} \sum_{j \notin \mathcal{I}_k} \log \left[c \left(\boldsymbol{F}_{n,\perp,(-\mathcal{I}_k)}(\boldsymbol{X}_j); \boldsymbol{\theta} \right) \right]$,
- $\boldsymbol{F}_{n,\perp,(-\mathcal{I}_k)}(x_1, \ldots, x_d) = \left(F_{(-\mathcal{I}_k),1}(x_1), \ldots, F_{(-\mathcal{I}_k),d}(x_d) \right)$, where $F_{(-\mathcal{I}_k),l}$ is the $\frac{n-n_v}{n}$-rescaled empirical cdf of the l-th marginal, computed from the sample $\mathcal{X}_n$ excluding $\{\boldsymbol{X}_j : j \in \mathcal{I}_k\}$, for $l = 1, \ldots, d$.

In our simulation study, we used both fixed and data-driven choices for the number of folds:

$$K_n \in \left\{ 2, 5, 10, \left\lfloor \frac{\sqrt{n}}{\log(n)} \right\rfloor, \lfloor \sqrt{n} \rfloor, \left\lfloor \frac{n}{\log(n)} \right\rfloor \right\}.$$

Summary of MPLE-based Information Criteria

In the semiparametric setting of copula model selection, we consider the following four information criteria:

1. Naive Akaike Information Criterion: ${}^{p}\mathrm{AIC}$ **(not valid, easy to compute)**
2. Leave-One-Out Cross-Validation: ${}^{p}\mathrm{xv}_1$ **(valid, computationally expensive)**
3. Approximate Leave-One-Out Criterion: $\mathrm{xv}_{\mathrm{CIC}}$ **(valid, moderately expensive to compute)**
4. K_n-fold Cross-Validation: ${}^{p}\mathrm{xv}_{K_n}$ **(?, computationally expensive)**

In the study Jordanger and Tjøstheim (2014), the authors compared $\mathrm{xv}_{\mathrm{CIC}}$ and ${}^{p}\mathrm{AIC}$.

Setup of the Simulation Study

The simulation study is based on the following settings:

- We constrain our analysis to bivariate distributions ($d = 2$).
- We simulate true data-generating models from five widely used one-parameter copula families:

  Clayton, Gumbel, Joe, Frank, Gaussian.

- Each copula was parametrized using different values of Kendall's tau

  $$\tau \in \{0.05, 0.1, 0.15, 0.2, 0.25, 0.5, 0.75\}.$$

- Considered sample sizes $n \in \{100, 200, 400\}$.
- For each combination of the true model, τ and n, we conducted 1000 Monte Carlo replications.

During each MC replication, all five candidate copula families are fitted to the generated data using the MPLE procedure.

Evaluation Metric: Hit Rates

Objective: Measure how often a criterion successfully identifies the true data-generating copula c°.

Step 1: Model Selection

- In each replication m, the criterion assigns a numerical score $S_m(\text{crit}, c)$ to the fitted candidate family c.
- The selected model is the one that maximizes this score:

$$c_{m,\text{crit}}^{\text{selected}} = \arg\max_{c \in \mathcal{C}_{\text{cand}}} S_m(\text{crit}, c).$$

Step 2: Empirical Hit Rate (HR)

- We define an indicator $I_m(\text{crit}) = \mathbb{I}\left(c_{m,\text{crit}}^{\text{selected}} = c^\circ\right)$.
- The Hit Rate is the sample mean across $M = 1000$ replications:

$$\text{HR}(\text{crit}) = \frac{1}{M}\sum_{m=1}^{M} I_m(\text{crit}).$$

Evaluation Metric: Hit Rates

Step 3: Asymptotic Confidence Intervals

- Treating the indicators as Bernoulli trials, we use the CLT to construct 95% CIs:

$$\mathrm{HR}(\mathrm{crit}) \pm z_{0.975} \times \frac{s_{\mathrm{crit}}}{\sqrt{M}},$$

  where s_{crit} denotes the sample standard deviation of the indicator variables $\{I_m(\mathrm{crit})\}_{m=1}^{M}$.

→ *Purpose:* This allows us to rigorously determine if performance differences between criteria are statistically significant.

Results and Discussion: Identification of the True Copula

IC	Clayton	Gumbel	Joe	Frank	Gaussian
^{p}AIC	55.50 $\pm$3.080	13.20 $\pm$2.100	51.30 $\pm$3.100	32.80 $\pm$2.910	18.00 $\pm$2.380
${}^{p}\mathrm{xv}_1$	55.50 $\pm$3.080	13.20 $\pm$2.100	51.30 $\pm$3.100	32.80 $\pm$2.910	18.00 $\pm$2.380
$\mathrm{xv}_{\mathrm{CIC}}$	55.70 $\pm$3.080	10.60 $\pm$1.910	53.60 $\pm$3.090	29.20 $\pm$2.820	18.40 $\pm$2.400
$K = 2$	40.00 $\pm$3.040	13.80 $\pm$2.140	43.90 $\pm$3.080	23.80 $\pm$2.640	16.20 $\pm$2.280
$K = 5$	38.50 $\pm$3.020	18.00 $\pm$2.380	48.20 $\pm$3.100	22.30 $\pm$2.580	13.80 $\pm$2.140
$K = 10$	38.80 $\pm$3.020	21.50 $\pm$2.550	50.60 $\pm$3.100	25.90 $\pm$2.720	13.90 $\pm$2.150
$K_n = \left\lfloor \frac{n}{\log(n)} \right\rfloor$	36.70 $\pm$2.990	24.40 $\pm$2.660	52.10 $\pm$3.100	24.60 $\pm$2.670	15.40 $\pm$2.240
$K_n = \sqrt{n}$	38.60 $\pm$3.020	22.40 $\pm$2.590	50.90 $\pm$3.100	24.50 $\pm$2.670	15.40 $\pm$2.240
$K_n = \left\lfloor \frac{\sqrt{n}}{\log(n)} \right\rfloor$	39.90 $\pm$3.040	15.30 $\pm$2.230	43.10 $\pm$3.070	23.70 $\pm$2.640	14.30 $\pm$2.170

Table 2: Hit rates ($n = 200$, $\tau = 0.05$), with 95 % confidence intervals (all values multiplied by 100).

Results and Discussion: Identification of the True Copula

IC	Clayton	Gumbel	Joe	Frank	Gaussian
^{p}AIC	97.90 $\pm$0.890	68.70 $\pm$2.880	86.40 $\pm$2.130	75.50 $\pm$2.670	67.80 $\pm$2.900
${}^{p}\mathrm{xv}_1$	97.90 $\pm$0.890	68.70 $\pm$2.880	86.40 $\pm$2.130	75.50 $\pm$2.670	67.80 $\pm$2.900
$\mathrm{xv}_{\mathrm{CIC}}$	96.10 $\pm$1.200	65.90 $\pm$2.940	88.60 $\pm$1.970	77.40 $\pm$2.590	67.00 $\pm$2.920
$K = 2$	96.00 $\pm$1.220	64.70 $\pm$2.960	83.20 $\pm$2.320	69.50 $\pm$2.860	61.90 $\pm$3.010
$K = 5$	97.70 $\pm$0.930	68.20 $\pm$2.890	85.50 $\pm$2.180	75.80 $\pm$2.660	67.50 $\pm$2.900
$K = 10$	97.60 $\pm$0.950	69.40 $\pm$2.860	84.80 $\pm$2.230	75.70 $\pm$2.660	66.90 $\pm$2.920
$K_n = \left\lfloor \frac{n}{\log(n)} \right\rfloor$	97.70 $\pm$0.930	69.20 $\pm$2.860	85.70 $\pm$2.170	77.00 $\pm$2.610	66.90 $\pm$2.920
$K_n = \sqrt{n}$	98.00 $\pm$0.870	68.70 $\pm$2.880	86.20 $\pm$2.140	77.40 $\pm$2.590	67.50 $\pm$2.900
$K_n = \left\lfloor \frac{\sqrt{n}}{\log(n)} \right\rfloor$	97.20 $\pm$1.020	67.90 $\pm$2.900	85.20 $\pm$2.200	75.60 $\pm$2.660	64.90 $\pm$2.960

Table 3: Hit rates ($n = 400$, $\tau = 0.25$), with 95 % confidence intervals (all values multiplied by 100).

Coincidence Percentages

Objective: Quantify how often two distinct criteria (crit_1 and crit_2) agree, regardless of whether their choice is the true copula model.

Step 1: The Coincidence Indicator

- In replication m, we check if both criteria choose the exact same candidate copula family:

$$J_m(\text{crit}_1, \text{crit}_2) = \mathbb{I}\left(c^{\text{selected}}_{m,\text{crit}_1} = c^{\text{selected}}_{m,\text{crit}_2}\right).$$

Step 2: Empirical Coincidence Percentage (CP)

- The CP is the sample mean of these agreements across all M replications:

$$\text{CP}(\text{crit}_1, \text{crit}_2) = \frac{1}{M}\sum_{m=1}^{M} J_m(\text{crit}_1, \text{crit}_2).$$

Results and Discussion: Criteria Agreement

	$\tau = 0.05$	$\tau = 0.1$	$\tau = 0.15$	$\tau = 0.2$	$\tau = 0.25$
100	99.74	99.92	99.90	99.96	99.94
200	99.92	99.98	99.98	99.98	99.98
400	100.00	99.98	100.00	100.00	100.00

Table 4: Coincidence of ${}^{p}\mathrm{AIC}$ and ${}^{p}\mathrm{xv}_1$ (all values multiplied by 100).

	$\tau = 0.05$	$\tau = 0.1$	$\tau = 0.15$	$\tau = 0.2$	$\tau = 0.25$
100	79.26	86.46	89.10	89.34	90.34
200	86.88	91.38	92.90	94.10	94.72
400	92.52	95.46	96.38	96.60	97.26

Table 5: Coincidence of ${}^{p}\mathrm{AIC}$ and $\mathrm{xv}_{\mathrm{CIC}}$ (all values multiplied by 100).

Summary: Semiparametric Model Selection

- **The Paradox of ^{p}AIC:** Despite lacking theoretical justification, it performs almost indistinguishably from the rigorously leave-one-out cross-validation ${}^{p}\mathrm{xv}_1$.
- **Performance of the $\mathrm{xv}_{\mathrm{CIC}}$:**
  - Serves as an excellent asymptotic approximation of ${}^{p}\mathrm{xv}_1$ in large samples.
  - Deviates in small samples because its complex bias-correction terms introduce finite-sample variation.
- **Evaluation of K_n-Fold Cross-Validation ${}^{p}\mathrm{xv}_{K_n}$:**
  - **Goal:** Employ data-driven folds to mimic the asymptotic properties of ${}^{p}\mathrm{xv}_1$ while reducing its computational cost.
  - **Conclusion:** Whether using fixed or dynamic choices for K_n, ${}^{p}\mathrm{xv}_{K_n}$ does **not** yield a statistically superior criterion compared to existing methods.

Conclusion: Guidelines for Copula Model Selection

Step 1: The Marginal Assumption

- *Margins known (e.g., via domain experts)?* $\rightarrow$ Fully Parametric Strategy (Two-Stage MLE).
- *Margins unknown?* $\rightarrow$ Semiparametric Strategy (MPLE).

Step 2: Fully Parametric Criteria ($\mathrm{AIC}_{\mathrm{2MLE}}$ vs. $\mathrm{CIC}_{\mathrm{2MLE}}$)

- **Large samples:** $\mathrm{AIC}_{\mathrm{2MLE}}$ is safe to use even when the true copula is misspecified. It is theoretically justified and performs identically to $\mathrm{CIC}_{\mathrm{2MLE}}$.
- **Small samples:** Exercise caution with $\mathrm{CIC}_{\mathrm{2MLE}}$. Its asymptotic bias-correction terms can become unstable.

Step 3: Semiparametric Criteria (${}^{p}\mathrm{AIC}$ vs. $\mathrm{xv}_{\mathrm{CIC}}$)

- In contrast to the parametric case, ${}^{p}\mathrm{AIC}$ completely lacks theoretical justification.
- Despite this, ${}^{p}\mathrm{AIC}$ empirically matches the performance of exact cross-validation ${}^{p}\mathrm{xv}_1$.

Thank you for your attention!

FULL SIMULATION STUDY

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